Dust Particles and Gas+Dust Self-Gravity in the AthenaK Shearing Box: I. The Dust-Module Merge (Phase 4a), the Shear-Periodic Deposit Fold, the Ideal-Gas Energy Update and the Particle Timestep under Shear (Phase 4b, i–iii)

`athenak-multigrid` tree, branch `dust-multigrid`

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Abstract

This document opens the *dust track* of the self-gravity program: Lagrangian dust particles in the shearing-box code with multigrid self-gravity on refined meshes, toward the gas+dust gravito-turbulent boxes of Baehr, Zhu & Yang (2022) with the radial pressure gradient they omitted. It records what the code does on test problems, phase by phase; the mechanisms are in the companion `dust_selfgravity_implementation.pdf`.

Phase 4a merges the sibling fork `athenak-dust` — particle-mesh dust with Benítez-Llambay–Krapp–Pessah implicit drag inside the Krapp et al. (2024) IMEX integrator, validated in its own repository — into the multigrid tree, and asks two questions: is the dust module unchanged by the merge, and is the gravity machinery unchanged by the dust module? Both answers are yes to every printed digit. On the merged tree the dust fork's own acceptance numbers reproduce exactly: the N-species damping ladder converges at second order in Δt with the reference errors 1.3529×10^{-7}, 3.4627×10^{-8}, 8.7724×10^{-9}, 2.2075×10^{-9} and total momentum conserved to 10^{-16}; the multi-species Nakagawa–Sekiya–Hayashi drift equilibrium holds to 10^{-16} over ten orbits; the damped dusty sound wave converges toward second order in resolution; the ion-neutral C-shock under `imex2+` is bit-identical to its baseline. The gravity solver's documented statics reproduce exactly (slab sheet error 1.203×10^{-12}, Tomida & Stone contraction 0.125, boundary-annuli shearing wave 5.950×10^{-3} / 2.057×10^{-3}), bit-identically at 4 ranks. The one defect found — a multi-rank deadlock of the additive deposit exchange against the tree's rank-packed messaging — is fixed and ranks 1, 2 and 4 agree to all digits. Finally the gas-only stratified gravito-turbulence box evolved with `rk2` and with `imex2+` (the integrator the dust module requires) agrees in every history column to between 2×10^{-6} and 5×10^{-3} at $t = 3\,\Omega^{-1}$, the expected spread between two second-order integrators.

A **second merge** the same day brought in the dust fork's latest seven commits. The whole battery reproduces on the result to every digit (the gas-only run bit for bit), and the fork's new predictor–corrector coupling under plain `rk2` (`coupling = pc2`) is exercised for the first time here: second order in the damping ladder (orders 2.23, 2.10, 2.04) and in the dusty wave (1.85, 1.98, 2.00), NSH to round-off, rank-independent at 1, 2 and 4 ranks — the integrator freedom that lets dust run under the same `rk2` as every gravity result before it.

Phase 4b (i) adds the conservative azimuthal remap of ghost deposits across the shear-periodic radial faces, the first of the gaps the dust module carried. A particle-free unit test certifies the new fold: the plain exchange delivers nothing across the faces, the fold is exact at integer shifts (3.5×10^{-16}) including the overlap multiplicity of corner rows, mass is conserved to 10^{-15}, nothing leaks outside the receiving strips, the remap error converges at second order

(donor cell), second/third (PLM), third (PPMX) in a fixed-phase scan, and 1, 2 and 4 ranks agree to every digit. The 3D shear-periodic NSH equilibrium still holds to round-off; a deterministic azimuthally structured dust distribution with back-reaction agrees between serial and 4-rank runs to 10^{-14} over an orbit, and shows the old unsheared exchange was a 1–7% error in the gas kinetic energy while the remap order matters only at 10^{-7}.

Phase 4b (ii) removes the isothermal restriction on back-reaction. With every drag kick booking its kinetic-energy change, the ideal-gas damping test keeps the momentum dynamics identical to the isothermal reference and changes the internal energy by 1% (IMEX) / 2% (PC2) of the dissipated energy at a step equal to the stiffest stopping time — the low-storage RK combination gap, converging with the step; the frictional-heating option conserves $E_{\rm gas} + K_{\rm dust}$ to 4–6% of the dissipated energy at that step, first order in Δt; and on the 3D shear-periodic NSH state the heating rate matches the analytic dissipation to 1.5%, 0.7%, 0.4% as the step is halved twice, with the equilibrium itself at round-off in both modes.

Phase 4b (iii) drops the background shear from the particle transport limit and replaces it by a MeshBlock-crossing guard on the total velocity. In an $L_x = 8$ shear-periodic box the ten-orbit NSH run takes 4588 cycles instead of 24043 (5.2×; the step is now the gas limit), at round-off in every drift velocity, serially, at 4 ranks and under PC2; the guard reproduces its formula to four digits when its safety fraction is varied, and with thin MeshBlocks it keeps a run clean that aborts on the first cycle without it. A structured run is rank-independent to 10^{-14} at the larger step. The scan the larger step made necessary shows the IMEX coupling's own truncation error at $\Delta t/t_{s,\rm min} = 1.4$: 1% in the gas kinetic energy in the quiet phase, 26% during the run's streaming burst, converging at second order — an error the legacy limit had been hiding in wide boxes by forcing $\Delta t \approx t_{s,\rm min}/4$ — while PC2 at its `rk2` step sits within 2% of the converged value.

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1 Plan and scope

1.1 The program

The science target is the interplay of gas gravitational instability and dust concentration: 3D stratified, cooling, gravito-turbulent shearing boxes with superparticles (Baehr, Zhu & Yang 2022: $512^2 \times 256$, $\beta = 10$ cooling with an irradiation floor, $\gamma = 5/3$, $\mathrm{St} = 0.1$, 1, 10, $Z = 0.01$, $Q_0 = 1.02$), extended by the radial pressure gradient that admits the streaming instability, and eventually adaptive zoom on the dust clumps — the configuration their limitations section names. The route (implementation document, §1.2): Phase 4a merges the dust module; 4b adds what the 3D shearing box needs from it (shear-remapped additive deposits, ideal-gas back-reaction, a relaxed particle timestep); 4c adds dust self-gravity through the existing Poisson solve; 4d builds the science configuration; Phase 5 takes particles onto refined meshes.

1.2 What Phase 4a must establish

1. **Dust fidelity.** The dust fork ships its validation in `../athenak_dust_tests/` (notebooks with printed reference values). On the merged tree, those references must reproduce: the damping ladder (second order in Δt, momentum to round-off), the NSH drift equilibrium (round-off over ten orbits), the damped dusty wave (spatial convergence), the random-placement and single-block layout checks, and the regression of the ion-neutral C-shock that shares the `imex2+` coefficient code.
2. **Gravity regression.** Three static multigrid tests whose numbers are documented in the multigrid validation document: the slab-open sheet test, the Tomida & Stone §4.1 contraction, and the boundary-annuli shearing wave including its zero-phase bit-identity with the periodic twin.
3. **Rank independence.** Dust and gravity at 1, 2 and 4 MPI ranks.
4. **Integrator.** The dust module's IMEX coupling requires `imex2+`; all gravity validation so far used `rk2`. A gas-only stratified gravito-turbulence box under both integrators, compared column by column.
5. **The second merge** (§4). After catching up with the dust fork: the battery again, and the new PC2/hybrid couplings under `rk2`.
6. **Phase 4b (i)** (§5). The shear-periodic deposit fold: exactness, convergence, conservation, rank independence, and its effect on a structured 3D run.
7. **Phase 4b (ii)** (§6). The ideal-gas energy update: momentum dynamics unchanged, energy budgets of both modes versus the step, the heating rate against the analytic dissipation.

2 Test setup

2.1 Builds

Serial and MPI builds in the tree's `build/` and `build-mpi/`, both with `-D PROBLEM=built_in_pgens -D Athena_ENABLE_FFT=ON`, the MPI one with `-D Athena_ENABLE_MPI=ON` (Open MPI via Homebrew). Everything below was first run on a scratch clone at the same commits and then rerun with the tree's own binaries; the two sets of results are identical, including the gravito-turbulence histories bit for bit.

2.2 Inputs and commands

Dust inputs come from the dust-tests repository (paths relative to the directory holding both trees):

```
# damping ladder (tlim = 1, so dt = 1/ncycle)
for c in 0.4 0.2 0.1 0.05; do
  athena -i athenak_dust_tests/inputs/dust_damping.athinput particles/ppc=3 time/
    cfl_number=$c
done
# layout variants
athena -i .../dust_damping.athinput particles/ppc=4 problem/lattice=false
athena -i .../dust_damping.athinput particles/ppc=4 problem/lattice=false dust/deposit=
    ngp
athena -i .../dust_damping.athinput particles/ppc=3 meshblock/nx1=32 meshblock/nx2=32
# NSH equilibrium, dusty wave, C-shock regression
athena -i .../dust_nsh.athinput
athena -i .../dusty_wave.athinput mesh/nx1=$n meshblock/nx1=$((n/2))     # n = 32..256
athena -i inputs/ion-neutral/perp_cshock.athinput time/integrator=imex2+ time/nlim=500
# rank independence (drifting particles cross MeshBlock and rank boundaries)
mpirun -np $np athena -i .../dust_damping.athinput particles/ppc=3 problem/v0=0.5 dust/
    taus_3=0.5
# gravity statics
athena -i inputs/tests/mg_poisson_slab.athinput
athena -i inputs/tests/ts41_sin3.athinput
athena -i inputs/tests/mg_poisson_shear_bndry.athinput problem/profile=shwave problem/
    time0=0.37
# integrator twin (gas only, uniform grid)
athena -i inputs/shearing_box/gravito_turb_amr_test.athinput mesh_refinement/refinement=
    none \
       time/tlim=3.0 [time/integrator=imex2+ time/cfl_number=0.3]
```

Phase 4b (iii) (§7; the wide box is the tracked 3D input with $x_1 \in [-4, 4]$ at the same cell size):

```
WIDE="mesh/nx1=64 mesh/x1min=-4.0 mesh/x1max=4.0 meshblock/nx1=32"
IN=inputs/dust/dust_nsh_shear3d.athinput
athena -i $IN $WIDE dust/dt_transport=full            # legacy limit, 10 orbits
athena -i $IN $WIDE dust/dt_transport=relative        # default; also mpirun -np 4, and
                                                      # dust/coupling=pc2 time/integrator
    =rk2 time/cfl_number=0.3
# structured state, one orbit, both limits, serial and 4 ranks, IMEX and PC2
athena -i $IN $WIDE problem/mass_mod_amp=0.5 problem/check_equilibrium=false \
       time/tlim=6.283185307179586 dust/dt_transport={full,relative}
# guard scaling (three cycles) and the thin-block abort demonstration (one orbit)
athena -i $IN $WIDE dust/dt_block_safety={0.1,0.2,0.4} time/nlim=3 time/ndiag=1
athena -i $IN $WIDE meshblock/nx2=4 dust/dt_block_safety={0.5,4.0} time/tlim
    =6.283185307179586
```

Each dust problem generator writes a `<basename>-errs.dat` line per run; the gravity statics print their metrics to standard output. All result files are archived under `validation/run/dust4a/` (see its `README.txt`), which the figure scripts `figures/plot_dust4a_*.jl` read.

3 Results

3.1 Dust fidelity

Damping ladder (Krapp et al. 2024 §3.1). Uniform gas at rest coupled to three dust species, $t_s = (0.01, 0.1, 1.0)$, $\epsilon_s = 1/3$ each, one particle of every species at every cell center so the discrete update per cell *is* the integrator's map of the ODE system. Errors are against a fine-step RK4 reference; the first run has $\Delta t > t_{s,1}$ (stiff regime).

Δt	err_u	err_{v1}	err_{v2}	err_{v3}	$\lvert\Delta P_{\mathrm{tot}}\rvert$
1.235×10^{-2}	1.3529×10^{-7}	1.3711×10^{-7}	1.4641×10^{-7}	6.8939×10^{-7}	1.6×10^{-16}
6.211×10^{-3}	3.4627×10^{-8}	3.5096×10^{-8}	3.7325×10^{-8}	1.7630×10^{-7}	7.5×10^{-17}
3.115×10^{-3}	8.7724×10^{-9}	8.8920×10^{-9}	9.4213×10^{-9}	4.4631×10^{-8}	2.0×10^{-16}
1.558×10^{-3}	2.2075×10^{-9}	2.2377×10^{-9}	2.3649×10^{-9}	1.1225×10^{-8}	1.7×10^{-16}
order of err_u	1.98, 1.99, 1.99				
reference (dust tests, 2026-07-14)	$\mathrm{err}_u = 1.3529 \times 10^{-7}$, 3.4627×10^{-8}, 8.7724×10^{-9}, 2.2075×10^{-9}; $\lvert\Delta P\rvert \sim 10^{-16}$				

Every reference digit reproduces (Fig. 1). The layout variants: random positions with TSC and with NGP deposit give $\mathrm{err}_u = 5.29 \times 10^{-6}$ and 5.77×10^{-6} (the 10^{-5}-level departure from the homogeneous ODE is physical: density fluctuations couple to pressure) with momentum to 3.8×10^{-16}; the single-MeshBlock lattice run reproduces the four-block ladder row to all digits, which is the halo-exchange exactness check.

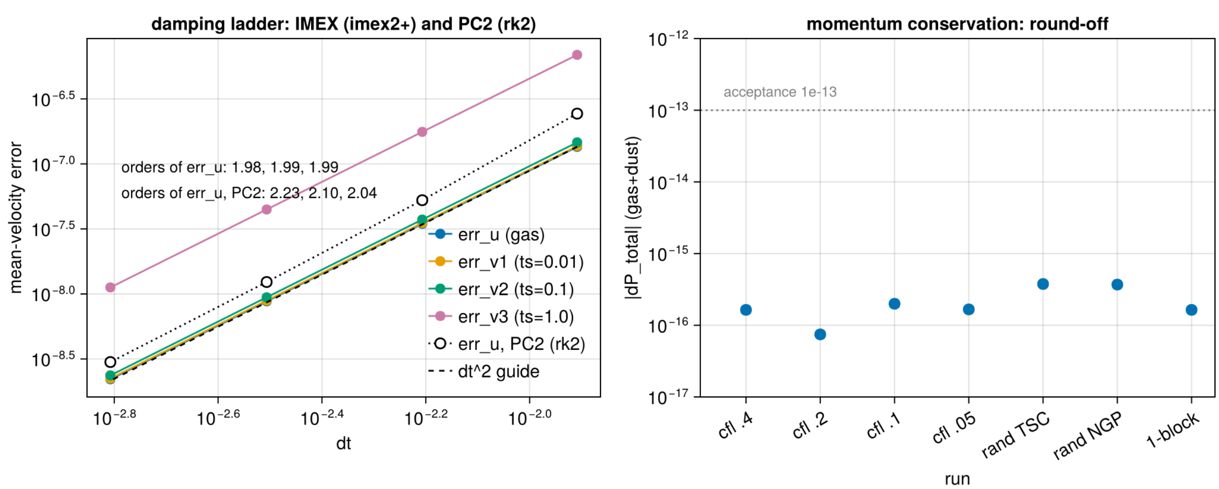

Figure 1: The damping ladder on the merged tree: second-order convergence of all four mean-velocity errors in Δt under the IMEX coupling (filled), the gas error under the PC2 coupling with `rk2` after the second merge (open, §4), and the total gas+dust momentum change of every IMEX run against the 10^{-13} acceptance line (right).

NSH drift equilibrium (2D r–z, ten orbits). Gas plus three species ($t_s = 0.01, 0.1, 1.0$; $\epsilon = 0.2, 0.5, 0.3$) started at the exact multi-species steady drift solution, with the gas forced by the pressure-gradient mimic $2\Omega\eta v_K$. After 4495 cycles every deviation is at round-off:

	$\lvert\delta u_x\rvert$	$\lvert\delta u_\phi\rvert$	$\lvert\delta v_{x,1}\rvert$	$\lvert\delta v_{\phi,1}\rvert$	$\lvert\delta v_{x,2}\rvert$	$\lvert\delta v_{\phi,2}\rvert$	$\lvert\delta v_{x,3}\rvert$	$\lvert\delta v_{\phi,3}\rvert$
serial	1.3×10^{-16}	6.2×10^{-17}	6.9×10^{-17}	5.9×10^{-17}	1.2×10^{-16}	2.4×10^{-17}	9.4×10^{-17}	4.3×10^{-17}
4 ranks	1.1×10^{-16}	9.0×10^{-17}	1.2×10^{-16}	2.8×10^{-17}	1.3×10^{-16}	1.0×10^{-17}	9.0×10^{-17}	3.3×10^{-17}

This is the test that certifies the rotation/forcing kicks and the implicit drag are composed unsplit inside the IMEX stages: any splitting error would show as secular drift.

Damped dusty sound wave (Laibe & Price 2011; Krapp et al. §3.2). The exact complex eigenmode of the linearized gas–dust system, L_1 errors of density and velocity against the analytic damped wave at $t = 2$:

N_x	cycles	$L_1(\delta\rho)$	$L_1(\delta v_x)$	order
32	320	1.836×10^{-8}	1.351×10^{-8}	—
64	321	6.896×10^{-9}	5.124×10^{-9}	1.41
128	641	1.837×10^{-9}	1.377×10^{-9}	1.91
256	1281	4.765×10^{-10}	3.572×10^{-10}	1.95

The code's own root of the dispersion relation, $\omega = 4.529763 - 0.492158\,i$, is printed identically in every run. The order approaches two from below; the $32 \to 64$ step is polluted by the fixed-`cfl` time step at the coarsest resolution (the same cycle count at 32 and 64 cells), as the dust-tests notebook also observes.

3.2 Regressions

The ion-neutral C-shock (`inputs/ion-neutral/perp_cshock.athinput`, `integrator = imex2+`, 500 cycles) is the direct consumer of the refactored `imex2+` coefficient code: RMS-$L_1 = 2.071216\times10^{-3}$, bit-identical to the dust fork's baseline. The Sod tube runs through the untouched hydro task list.

3.3 Rank independence, and the deadlock

The damping problem with drifting particles (`v0 = 0.5`, $t_{s,3} = 0.5$, three lattice species) sends particles across MeshBlock and rank boundaries many times per run and exercises per-stage migration, the additive deposit exchange, the u^* ghost fill and the back-reaction exchange.

The first attempt at 4 ranks **hung at cycle 0**. The cause was in the merge itself: the dust fork's additive deposit exchange posted per-neighbor MPI messages against the base class of July's upstream, while this tree carries upstream #775, whose base-class receive posting is rank-packed (one aggregated message per peer rank). The sends had no receiver. After moving the exchange onto the rank-packed path (implementation document §4.4) with the serial path untouched (the damping ladder is bit-identical before and after the edit):

ranks	err_u	err_{v1}	err_{v2}	err_{v3}	$\lvert\Delta P_{\mathrm{tot}}\rvert$
1	1.158578×10^{-5}	1.187422×10^{-5}	1.490167×10^{-5}	6.153323×10^{-5}	5.2×10^{-15}
2	1.158578×10^{-5}	1.187422×10^{-5}	1.490167×10^{-5}	6.153323×10^{-5}	3.7×10^{-15}
4	1.158578×10^{-5}	1.187422×10^{-5}	1.490167×10^{-5}	6.153323×10^{-5}	3.6×10^{-15}
reference	1.158578×10^{-5}	(dust tests: “1-rank and 4-rank identical”)			

The tiny spread in $\lvert\Delta P\rvert$ is the reduction order of the diagnostic itself. The NSH run at 4 ranks is in the table of §3.1.

3.4 Gravity regression

test	metric	measured	documented
`mg_poisson_slab`, sheet profile (absolute, no mean subtraction)	max rel / L_2 rel	1.203474×10^{-12} / 3.078736×10^{-13}	1.2×10^{-12}
`ts41_sin3` (Tomida & Stone §4.1)	defect ratio per V-cycle	0.1254, 0.1249, 0.1249, 0.1249	0.125
`mg_poisson_shear_bndry`, shwave, $t_0 = 0.37$	max rel / L_2 rel	5.950118×10^{-3} / 2.057024×10^{-3}	5.95×10^{-3} / 2.057×10^{-3}
same, $t_0 = 0$, shear-periodic vs periodic-x_1 twin	both runs	5.506611×10^{-3} / 1.281655×10^{-3}, identical	bit-identical

At 4 ranks the slab and boundary-annuli solves reproduce the serial digits exactly, as documented for Phases 2b and 3.

The dust module is inert without a `<dust>` block by construction (the hydro and gravity paths do not see it), and these numbers confirm the shared infrastructure it touched — driver coefficients, boundary-values base class, output tables, physics registration — left the solver untouched.

3.5 The integrator twin

The Phase-3 small stratified gravito-turbulence box (`inputs/shearing_box/gravito_turb_amr_test.athinput`: $16H \times 16H \times 12H$, $64 \times 64 \times 48$, $\beta = 10$, slab-open multigrid gravity, refinement switched off) evolved to $t = 3\,\Omega^{-1}$ with `rk2` and with `imex2+`, both at `cfl_number = 0.3`. Relative differences of the history columns at $t = 3$:

	mass	$E_{\rm tot}$	$E_{\rm int}$	$\langle \rho\, c_s \rangle$	$\langle \rho\, w_{\rm grv} \rangle$	$\langle \rho\, w_{\rm rey} \rangle$	$\langle \rho\, \delta v^2 \rangle$
$\lvert$imex2+ $-$ rk2$\rvert/\lvert$rk2$\rvert$	2.0×10^{-6}	7.0×10^{-5}	6.3×10^{-5}	3.5×10^{-5}	5.3×10^{-3}	3.2×10^{-4}	3.2×10^{-4}

Two different second-order integrators (three explicit stages of which the first advances by $\gamma \Delta t \approx 1.71 \Delta t$, versus two) on a chaotic flow: agreement at 10^{-4}–10^{-3} in the bulk quantities, with the small gravitational stress — a difference of large terms at this early time — the loosest at 5×10^{-3} (Fig. 2). The spikes of the relative difference of the stress columns at $t \approx 1.75$ and 2.5 in the figure are zero crossings of the stresses themselves (top-left panel), where a relative measure carries no information. This is the level at which `rk2`-validated gravity results transfer to dust runs; the full-size SC14 statistics under `imex2+` are a cluster item and will be attached to the next β-scan submission.

4 The second merge: regression, and the PC2/hybrid couplings

The dust fork's `dust` branch had moved on by seven commits (to `1f3e1ed9`) while the first merge was based on `d6de1296`; commit `78320aee` merges them (implementation document §4.5). They bring the fork's own rank-packed deposit exchange (taken in place of ours), CPU particle-mesh optimizations, and two new coupling modes selectable with `<dust>/coupling`: `pc2`, a per-cycle predictor–corrector particle update under `rk2`, and `hybrid`, which switches between PC2 and a first-order split backward-Euler branch on stiffness measures $\zeta = \max \Delta t / t_s$ and $\chi = \max \Delta t \rho_d / (\rho_g t_s)$.

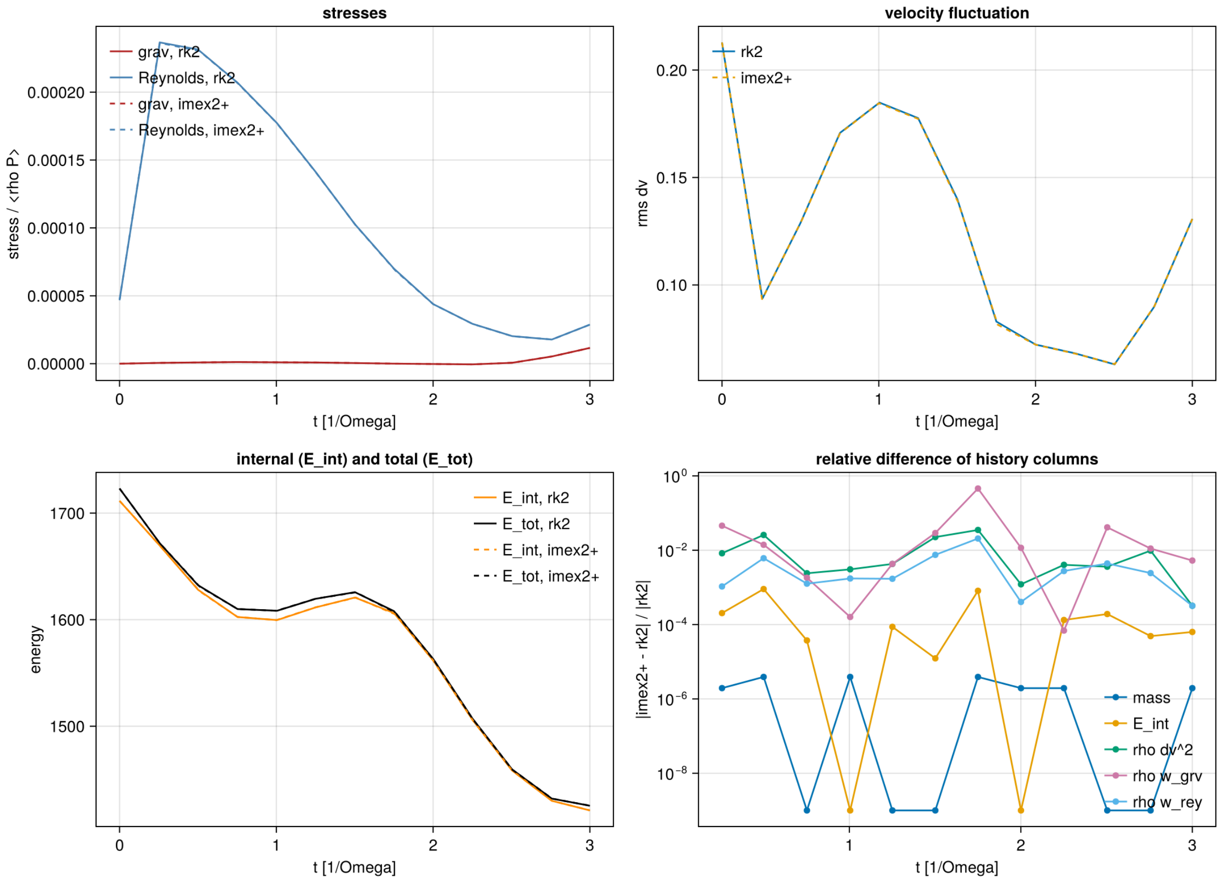

Figure 2: The gas-only gravito-turbulence box under `rk2` (solid) and `imex2+` (dashed): stresses normalized by $\langle \rho P \rangle$, rms velocity fluctuation, internal and total energy, and the relative difference of the history columns.

4.1 Regression

Every table of §3 was rerun on the rebuilt serial and MPI binaries. The damping ladder, its three layout variants, the dusty-wave ladder, the C-shock, the drifting-particle test at 1, 2 and 4 ranks, and all multigrid statics (serial and 4 ranks) reproduce the printed digits above exactly; the NSH deviations move within their 10^{-16} round-off (the new non-atomic deposit changes the summation order); and the gas-only gravito-turbulence run is **bit-identical** to the pre-merge history, as it must be — the merge touched nothing on the gas path.

4.2 PC2 coupling under `rk2`

The inputs are the same test inputs with `integrator = rk2` and `coupling = pc2` added to the `<dust>` block (archived under `run/dust4a/merge2/`; the key has to be in the file because the command line cannot add parameters). Damping ladder:

Δt	err_u	err_{v1}	err_{v2}	err_{v3}	$\lvert\Delta P_{\mathrm{tot}}\rvert$
1.235×10^{-2}	2.4349×10^{-7}	2.2596×10^{-7}	2.4748×10^{-7}	1.2039×10^{-6}	2.4×10^{-16}
6.211×10^{-3}	5.2577×10^{-8}	4.8577×10^{-8}	5.3355×10^{-8}	2.5966×10^{-7}	4.5×10^{-17}
3.115×10^{-3}	1.2351×10^{-8}	1.1216×10^{-8}	1.2461×10^{-8}	6.0729×10^{-8}	5.0×10^{-17}
1.558×10^{-3}	2.9941×10^{-9}	2.7081×10^{-9}	3.0167×10^{-9}	1.4707×10^{-8}	5.1×10^{-17}
order of err_u	2.23, 2.10, 2.04	(IMEX ladder: 1.98, 1.99, 1.99)			

Second order, converging onto the Δt^2 slope from above, with errors about 1.4–1.8× the IMEX ones at equal Δt and momentum at round-off (Fig. 1, open markers). The dusty wave:

N_x	$L_1(\delta\rho)$	$L_1(\delta v_x)$	order	(IMEX order)
32	1.612×10^{-8}	1.182×10^{-8}	—	—
64	4.483×10^{-9}	3.310×10^{-9}	1.85	1.41
128	1.139×10^{-9}	8.404×10^{-10}	1.98	1.91
256	2.858×10^{-10}	2.109×10^{-10}	2.00	1.95

Cleaner second order than the IMEX ladder and smaller errors at every resolution. The NSH equilibrium under `pc2` holds to round-off over the ten orbits (3.2×10^{-16}, 1.3×10^{-16}, 3.1×10^{-16}, 2.1×10^{-16}, 3.6×10^{-16}, 2.4×10^{-16}, 1.0×10^{-17}, 1.7×10^{-16} for the eight components of §3.1). Rank independence with the drifting particles:

ranks	err_u	err_{v1}	err_{v2}	err_{v3}	$\lvert\Delta P_{\mathrm{tot}}\rvert$
1	6.805779×10^{-6}	5.812361×10^{-6}	7.380154×10^{-6}	3.360985×10^{-5}	1.5×10^{-14}
2	6.805779×10^{-6}	5.812361×10^{-6}	7.380154×10^{-6}	3.360985×10^{-5}	1.9×10^{-15}
4	6.805779×10^{-6}	5.812361×10^{-6}	7.380154×10^{-6}	3.360985×10^{-5}	3.9×10^{-15}

4.3 Hybrid coupling and the split-BE branch

Two runs characterize the fallback. The three-species NSH test under `coupling = hybrid` (automatic selection) reports `pc2_cycles=0 split_be_cycles=4494` with $\zeta = 0.549$: the $t_s = 0.01$ species puts $\Delta t/t_s$ just above the 0.5 entry threshold, so the run stays in the first-order branch throughout, and the equilibrium is no longer an exact fixed point — deviations of $1\text{–}12 \times 10^{-5}$ after ten orbits instead of round-off. This is the designed behavior, not a defect: the first-order splitting error of a stiff fallback, on a problem whose IMEX and PC2 answers are exact. The stiff damping case of the dust tests ($t_s = 10^{-4}, 10^{-3}, 2\times10^{-3}$, $\Delta t/t_s$ up to 125) under `hybrid` runs split-BE for all 81 cycles ($\zeta = 32.8$, $\chi = 12.6$) and lands on the exact relaxed state (errors $\leq 4.9\times10^{-16}$, momentum 1.1×10^{-16}), the L-stability that the branch exists for. For production runs in the resolved regime the recommendation is therefore `coupling = pc2`; `hybrid` is the safety net for configurations with stopping times below the timestep.

5 Phase 4b (i): the shear-periodic deposit fold

The mechanism is in the implementation document §5: the x_1 ghost slabs of the shear-face MeshBlocks are gathered into global y–z planes, remapped in y by $\mp y_{\rm sh}$ (the adjoint of the ghost-fill signs), and added into the active edge strips across the face, while the unsheared plain-periodic x_1 buffers of those blocks are skipped. Data: `run/dust4b/` (README there).

5.1 The unit test

`pgen_name = dust_deposit_shear` (`inputs/dust/dust_deposit_shear.athinput`: $32\times32\times8$, $16\times16\times8$ blocks so both x_1 columns are face blocks, `nghost = 3`, $q = 3/2$, $\Omega = 1$, $L_x = 1$). Three fields live only in the x_1 ghost slabs of the face blocks: a smooth periodic g on the slab rows within the block's own y/z range, a constant on the same rows, and a constant on the whole slab including the corner ghosts (whose expectation is the analytic overlap multiplicity).

case	$y_{\rm sh}/\Delta y$	max rel (g)	const	multipl.	leak	mass
plain pass only (`fold = false`)	0	0	0	0	0	(none delivered)
fold, `time0 = 0`	0	0	0	0	0	5.9×10^{-16}
fold, 3-cell integer shift	3	3.5×10^{-16}	0	0	0	5.9×10^{-16}
fold, generic phase, $4 = 2 = 1$ ranks	35.52	1.332091×10^{-3}	0	(frac.)	0	10^{-15}

The first row is the suppression check: with the fold switched off the receiving strips stay *exactly* zero, so no unsheared x_1 contribution survives (the first version of the direction table failed this row for the x_3x_1 corners; implementation document §5.4). Rows two and three are the exactness checks, including the multiplicity of overlapping corner rows. At a fractional shift the multiplicity field is a step profile and carries an $O(1)$ remap error at block boundaries, as expected; the smooth field's error is the remap interpolation error, which the last row shows to be identical at 1, 2 and 4 ranks to every printed digit.

Convergence. At a fixed fractional shift ($y_{\rm sh}/\Delta y = n_y/2 + 1/2$, so `eps` $= 1/2$ at every resolution; the generic-phase scan is confounded by the changing fraction):

n_y	DC L_1	DC max	PLM L_1	PLM max	PPMX L_1	PPMX max
32	1.86×10^{-3}	1.24×10^{-2}	2.69×10^{-4}	6.40×10^{-3}	1.11×10^{-4}	2.11×10^{-3}
64	4.66×10^{-4}	3.18×10^{-3}	3.89×10^{-5}	1.33×10^{-3}	9.16×10^{-6}	4.76×10^{-4}
128	1.17×10^{-4}	7.96×10^{-4}	5.33×10^{-6}	6.49×10^{-4}	9.90×10^{-7}	1.17×10^{-4}
256	2.91×10^{-5}	1.99×10^{-4}	6.21×10^{-7}	1.47×10^{-4}	9.78×10^{-8}	2.92×10^{-5}
order	2.00	2.0	2.8–3.1	$\sim$2	3.2–3.6	2.0

Donor cell is exactly second order (the half-cell shift is symmetric); PLM, the default (matching the u^* remap), is second order in the maximum norm and better than second in L_1; PPMX is third order in L_1 with a limiter-bound maximum (Fig. 3). Mass is conserved to $\leq 3 \times 10^{-15}$ and the leak is exactly zero in every run.

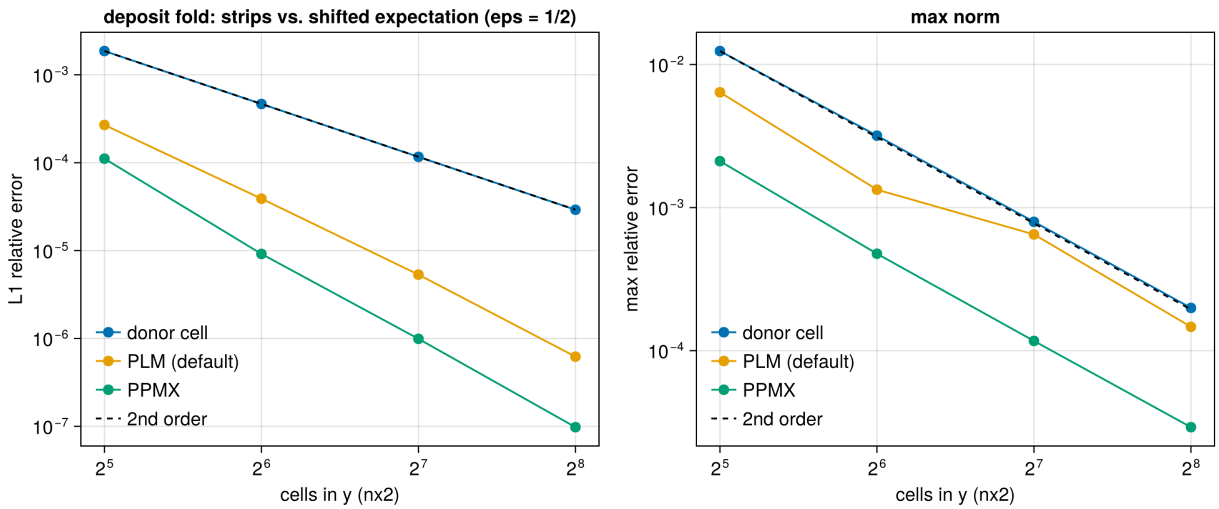

Figure 3: Convergence of the shear-periodic deposit fold at a fixed half-cell fractional shift, for the three remap orders.

5.2 Regression: the 3D shear-periodic NSH equilibrium

The new tracked input `inputs/dust/dust_nsh_shear3d.athinput` ($32 \times 32 \times 8$, $2 \times 2 \times 1$ blocks, ten orbits) is the azimuthally uniform state for which the old unsheared exchange was already exact:

	max deviation, before the fold	with the fold	PC2, with the fold	4 ranks
NSH, 3D shear-periodic	2.7×10^{-16}	2.6×10^{-16}	1.8×10^{-15}	1.1×10^{-16}

5.3 A structured run: rank independence and the size of the correction

Random particle placement in the NSH generator is rank-dependent by construction (it hashes tags to *local* block indices), which made a first serial-vs-MPI comparison look like a defect; with test particles the same runs agreed to 10^{-13}, and the cycle-0 dust density already differed. The deterministic alternative is the lattice with a 50% sinusoidal azimuthal modulation of the particle masses (`problem/mass_mod_amp = 0.5`, `check_equilibrium = false`), IMEX coupling with back-reaction, one orbit. Relative differences of the gas x-kinetic energy and of the species-1 radial-velocity sum:

t	serial vs 2 ranks	serial vs 4 ranks	fold vs. no fold	PLM vs. PPMX remap
0.81	10^{-14} / 7×10^{-15}	5×10^{-15} / 9×10^{-15}	4.2×10^{-2} / 2.7×10^{-2}	3.7×10^{-7} / 7.9×10^{-8}
3.20	1.2×10^{-14} / 5×10^{-15}	1.2×10^{-14} / 9×10^{-15}	9.7×10^{-3} / 5.8×10^{-3}	5.4×10^{-9} / 2.1×10^{-8}
4.80	6×10^{-15} / 7×10^{-15}	6×10^{-15} / 5×10^{-15}	6.8×10^{-2} / 3.9×10^{-2}	4.6×10^{-7} / 3.1×10^{-7}
6.28	6×10^{-15} / 5×10^{-15}	6×10^{-15} / 5×10^{-15}	5.4×10^{-3} / 3.6×10^{-3}	4.4×10^{-7} / 2.6×10^{-7}

Three conclusions. The fold is rank-independent to round-off on a structured field with back-reaction. The old unsheared exchange (`dust/shear_fold = false`) was a 1–7% error in the gas kinetic energy for a structured state, oscillating with the shear phase — not a small correction. And the remap order is a 10^{-7} effect on this run, so PLM is converged for the purpose.

6 Phase 4b (ii): the ideal-gas energy update

Mechanism: implementation document §6. Every drag kick on the gas momentum now books the matching kinetic-energy change (internal energy fixed under the kick); with `dust/drag_heating = true` the frictional dissipation of each particle kick is deposited as heat. Data: `run/dust4b/energy/`.

6.1 The damping test with an ideal gas

`inputs/dust/dust_damping_ideal.athinput`: the three-species damping problem of §3.1 with `eos = ideal`, $\gamma = 1.4$ and $p_{\rm gas} = 10^{-5}$ (comparable to the 1.4×10^{-5} of kinetic energy the drag dissipates, so the six-digit history resolves the budget; the first attempt with $p = 1$ could not), `dtmax` $= 1/95$ (the isothermal ladder's first step). The internal energy is $e = E - K_{\rm gas}$ from the hydro history, the dust kinetic energy $K_{\rm dust} = \sum_s \frac{1}{2}\epsilon_s\rho_0 V\bar{v}_s^2$ from the species velocity history (exact on the lattice).

Momentum dynamics. The four errors of the ODE test are identical between the default and heating modes to all printed digits (9.733252×10^{-8} etc. under IMEX, 1.660169×10^{-7} under PC2) — pressure and heat do not touch the velocities on a uniform state — and momentum is conserved to 10^{-16}.

Energy budgets versus the step. Errors relative to the dissipated kinetic energy, default mode ($|\Delta e|$) and heating mode ($|\Delta(E_{\rm gas}+K_{\rm dust})|$), with the order between successive rows in parentheses:

Δt	IMEX default	IMEX heating	PC2 default	PC2 heating
1/95	1.07×10^{-2}	3.81×10^{-2}	2.15×10^{-2}	5.89×10^{-2}
1/190	6.83×10^{-3} (0.65)	2.80×10^{-2} (0.44)	7.14×10^{-3} (1.59)	3.09×10^{-2} (0.93)
1/380	4.17×10^{-3} (0.71)	1.93×10^{-2} (0.54)	2.82×10^{-3} (1.34)	1.53×10^{-2} (1.01)
1/760	2.43×10^{-3} (0.78)	1.23×10^{-2} (0.65)	1.24×10^{-3} (1.18)	7.56×10^{-3} (1.02)

The default-mode residual is the register-combination gap $\gamma_0\gamma_1|\mathbf{K}_1|^2/2\rho$ per step (implementation document §6.3): the 1% at $\Delta t = 1/95$ is reproduced by that estimate with the stage-1 kick of the $t_s = 0.01$ species, and the sub-first-order IMEX slope is the saturation of that kick while $\Delta t \gtrsim t_s$ (the orders climb toward one as the step shrinks; PC2, with its RK2 weights, is already first order and better). The heating mode inherits the same gap on the dust side and converges at first order. Both are truncation errors of the tableau, not bookkeeping mistakes: the sign, size and Δt-scaling are the predicted ones.

6.2 The NSH state with an ideal gas

`inputs/dust/dust_nsh_shear3d_ideal.athinput` ($p = 1$, two orbits):

	max deviation of the drift equilibrium	$\dot{e}$ / analytic dissipation rate
default mode, `cfl 0.45`	4.6×10^{-15}	(no heat: e constant to 8×10^{-5})
heating, `cfl 0.45` ($\Delta t/t_{s,1} = 1.16$)	4.6×10^{-15}	1.0148
heating, `cfl 0.225`	—	1.0074
heating, `cfl 0.1125`	—	1.0037
heating, `cfl 0.45`, 4 ranks	6.7×10^{-16}	1.014777 (serial: 1.014777)

The analytic rate is $\sum_s \epsilon_s \rho_0 |\mathbf{u} - \mathbf{v}_s|^2/t_s$ from the equilibrium velocities. The heating rate converges to it at first order (the error halves with the step; the stiff $t_s = 0.01$ species dominates), the equilibrium is untouched by either mode, and 4 ranks reproduce the serial rate to every digit.

6.3 Regression

Isothermal runs cannot enter the new branches. The 2D damping ladder is bit-identical to the second-merge battery; the 2D and 3D NSH errors and the structured-run histories differ from their Phase-4b (i) values only in quantities at the 10^{-16} level (the deposit kernels changed shape and the compiler schedules the fused multiply-adds differently), and the 4-rank drifting-particle test reproduces the reference 1.158578×10^{-5} with the now four-channel exchange.

7 Phase 4b (iii): the particle timestep under shear

Mechanism: implementation document §7. The transport limit is now the peculiar velocity per cell (`dust/dt_transport = relative`, default) with a guard that no particle crosses more than `dt_block_safety` $= 0.5$ of a MeshBlock per stage on the total velocity $v_y - q\Omega x$; `dt_transport = full` is the legacy limit. Data: `run/dust4b/dt/`; figure 4.

7.1 Regression at the old step

The 2D damping ladder is bit-identical to Phase 4a (cycle counts 81, 161, 321, 642 and every error digit): the guard is inert without shear. In the $L_x = 1$ box of the tracked 3D input both particle limits lie above the gas limit (1.37×10^{-2} from c_s and $\Delta x_2 = 1/32$): legacy 2.1×10^{-2}, new guard 1.9×10^{-1}. The two modes therefore take the same 4588 steps and produce identical error lines (1.8×10^{-16}), PC2 its usual 6882 steps at 1.2×10^{-15} — which is also why the penalty had never shown in this test.

7.2 The wide box

The same input with $x_1 \in [-4, 4]$ at the same cell size (64×32×8, blocks 32×16×8, max $|q\Omega x| = 5.9$), ten orbits, uniform NSH state:

run	limit	Δt	cycles	CPU s	max NSH deviation
IMEX, serial	`full` (legacy)	2.60×10^{-3}	24043	455	8.8×10^{-16}
IMEX, serial	`relative`	1.37×10^{-2}	4588	94	1.3×10^{-15}
IMEX, 4 ranks	`relative`	1.37×10^{-2}	4588	30	1.1×10^{-16}
PC2/`rk2`, serial	`relative`	9.13×10^{-3}	6882	99	1.2×10^{-15}
IMEX ideal gas, 2 orbits	`relative`	1.37×10^{-2}	1082	—	5.4×10^{-15}

The legacy step is $0.5\,\Delta x_2/5.9$; the new step is the gas limit, the particle guard standing at $(0.5/1.707)\cdot 0.5/5.9 = 2.5\times10^{-2}$ above it. Every particle now hops MeshBlocks in y routinely (up to 4.4 cells per stage) and crosses the shear-periodic face with multi-cell azimuthal offsets; ten orbits pass without a halo violation, at round-off, on one rank and on four.

The guard. Lowering `dt_block_safety` until the guard is the active limit reproduces its formula $s/\beta_{\max}\cdot L_y^{\mathrm{mb}}/\max|v_y - q\Omega x|$ with $\beta_{\max} = 1.707$, $L_y^{\mathrm{mb}} = 0.5$ and $\max|v_y - q\Omega x| = 1.5\cdot 3.9375 + 0.027$:

s	0.1	0.2	0.4
formula	4.937×10^{-3}	9.873×10^{-3}	1.975×10^{-2}
measured Δt	4.936×10^{-3}	9.873×10^{-3}	1.370×10^{-2} (gas cap)

That the guard is needed, and not merely correct, is shown with thin MeshBlocks (`meshblock/nx2 = 4`, $L_y^{\mathrm{mb}} = 0.125$): at the gas step a particle near the radial edge moves $1.707\cdot 0.0137\cdot 5.9 = 0.14$ per stage, more than a block. With the guard the step drops to 6.12×10^{-3} and an orbit (1023 cycles) runs clean; with it disabled (`dt_block_safety = 4`, warned) the run aborts on the first cycle with the PM halo violation for a particle at $x = -3.81$, exactly the failure mode of §7.2 of the implementation document.

7.3 The structured run, and what the larger step costs

The azimuthally modulated state of §5 (`mass_mod_amp = 0.5`, back-reaction, one orbit) in the wide box. Rank independence first: the serial and 4-rank runs at the new step agree to 10^{-14} in every gas history column and to 5×10^{-14} in the species velocity sums, the same round-off as before at five times the step.

Between the two limits the histories differ visibly, and the difference is worth understanding rather than tabulating: for IMEX the gas x-kinetic energy at $\Delta t = 1.37\times10^{-2}$ differs from the legacy-step run by 5% at $t = 0.5$, 1% at $t = 3$ and 21% at $t = 6.28$, where the run goes through a streaming burst (figure 4, middle); for PC2 at its 9.1×10^{-3} the corresponding number is 1.5%. The step scan below (`<time>/dtmax`, the scratch input `run/dust4b/nsh3d_dtmax.athinput`) shows what this is:

gas x-KE	IMEX, $\Delta t =$ 1.37×10^{-2}	6.85×10^{-3}	3.43×10^{-3}	1.71×10^{-3}	orders
$t = 3$	1.40011×10^{-4}	1.38681×10^{-4}	1.38473×10^{-4}	1.38435×10^{-4}	2.7, 2.5
$t = 6.28$	2.6477×10^{-4}	2.2227×10^{-4}	2.1152×10^{-4}	2.0752×10^{-4}	2.0, 1.4
	PC2, $\Delta t =$ 9.1×10^{-3}	4.55×10^{-3}	2.28×10^{-3}		
$t = 3$	1.38623×10^{-4}	1.38550×10^{-4}	1.38496×10^{-4}		
$t = 6.28$	2.0991×10^{-4}	2.0835×10^{-4}	2.0658×10^{-4}		

The IMEX sequence converges (successive differences fall by 4–6× per halving; the order drops toward the end of the burst, where the state changes fastest), so the 21% is the IMEX coupling's truncation error at $\Delta t/t_{s,1} = 1.4$ — the stiffest species is under-resolved in time — and not an effect of the transport rule: the legacy limit had been supplying, by accident and only in wide boxes, a step of $t_{s,1}/4$. PC2 is closer to the converged value at every step (its 9.1×10^{-3} run is within 2% of the extrapolated IMEX limit, IMEX's own 2.6×10^{-3} run within 1.5%), consistent with the second-merge finding that PC2 is the production coupling in the resolved regime. The `gamma_switch` is inert here (it keys on the *largest* stopping time, $t_{s,3} = 1$). The practical rule: with `coupling = imex` and species stiffer than the gas step, cap the step with `<time>/dtmax` at a fraction of $t_{s,\min}$ if their dynamics matter; under PC2 the `rk2` step is adequate.

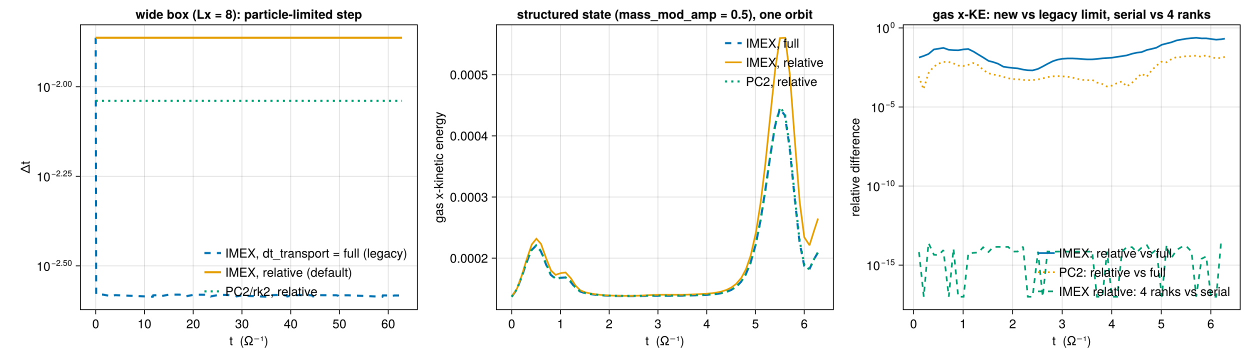

Figure 4: Phase 4b (iii). Left: the timestep of the wide-box NSH run under the legacy limit and the new default, IMEX and PC2. Middle: gas x-kinetic energy of the structured run over one orbit under both limits; the burst near $t = 5.5$ is where the IMEX truncation error at $\Delta t = 1.4\, t_{s,1}$ is largest. Right: relative differences of that history — new versus legacy step (truncation) and serial versus 4 ranks (round-off).

8 Summary

Phase 4a delivers the dust module inside the multigrid tree with both sides certified unchanged: the dust fork's acceptance numbers and the gravity solver's documented statics reproduce to every printed digit, serially and at 4 ranks, and the one merge defect (the additive deposit exchange against rank-packed messaging) is fixed with a rank-independence table to show for it. The integrator the dust module's IMEX coupling imposes, `imex2+`, tracks `rk2` on the stratified gravito-turbulent box at the level two second-order schemes should — and after the second merge that constraint is optional: the PC2 coupling runs under `rk2` at second order with the same conservation properties, and is the recommended production path in the resolved drag regime.

Phase 4b (i) closes the first gap Phase 4a left open: ghost deposits now cross the shear-periodic radial faces with a conservative azimuthal remap, certified exact at integer shifts, convergent otherwise, conservative, leak-free and rank-independent, and shown to matter at the percent level on structured states. Phase 4b (ii) makes back-reaction usable with an ideal gas: the energy bookkeeping is exact per kick, the residuals are the RK combination gap (a percent of the dissipated energy at a step equal to the stiffest stopping time, converging), and the optional frictional heating reproduces the analytic dissipation rate to first order. Phase 4b (iii) closes the third: the particle step no longer pays the full-shear-velocity penalty — 5.2× fewer cycles in an $L_x = 8$ box, with the equilibrium, rank independence and the MeshBlock guard each certified — and the scan it prompted fixes the operating rule for stiff species under IMEX (cap the step at a fraction of $t_{s,\min}$, or use PC2). Phase 4b is complete; Phase 4c adds the dust to the Poisson source.